

Overview and Terminology

In this Guide we address the full spectrum of spatial analysis and associated modeling techniques that are provided within currently available and widely used geographic information systems (GIS) and associated software. Collectively such techniques and tools are often now described as *geospatial analysis*, although we use the more common form, *spatial analysis*, in most of our discussions.

The term ‘GIS’ is widely attributed to Roger Tomlinson and colleagues, who used it in 1963 to describe their activities in building a digital natural resource inventory system for Canada (Tomlinson 1967, 1970). The history of the field has been charted in an edited volume by Foresman (1998) containing contributions by many of its early protagonists. A timeline of many of the formative influences upon the field is provided in [Longley et al. \(2015, p20\)](#). The research makes the unassailable point that the success of GIS as an area of activity has been driven by the success of its applications in solving real world problems.

In order to cover such a wide range of topics, this Guide has been divided into a number of main sections or chapters. These are then further subdivided, in part to identify distinct topics as closely as possible, facilitating the creation of a web site from the text of the Guide. Hyperlinks embedded within the document enable users of the web and PDF versions of this document to navigate around the Guide and to external sources of information, data, software, maps, and reading materials.

[Chapter 2](#) provides an introduction to spatial thinking, described by some as “spatial literacy”, and addresses the central issues and problems associated with spatial data that need to be considered in any analytical exercise. Readers who are already familiar with these concepts may wish to skip this chapter and move straight on to [Chapter 3](#).

Real-world problems and applications are typically governed by the organizational *practices* and *procedures* that prevail with respect to particular *places*. Not only are there wide differences in the volume and remit of data that the public sector collects about population characteristics in different parts of the world, but there are differences in the ways in which data are collected, assembled and disseminated (e.g. general purpose censuses versus statistical modeling of social surveys, property registers and tax payments). Data collected by the private sector, often as a result of the use of services that automatically gather information on events, locations and individuals, present a further challenge (see Chapter 9 for an extended discussion of this issue). There are also differences in the ways in which different data holdings can legally be merged and the purposes for which data may be used — particularly with regard to health and law enforcement data. Finally, there are geographical differences in the cost of geographically referenced data. Some organizations, such as the US Geological Survey, are bound by statute to limit charges for data to sundry costs such as media used for delivering data while others, such as most national mapping organizations in Europe, are required to exact much heavier charges in order to recoup much or all of the cost of data creation. Analysts may already be aware of these contextual considerations through local knowledge, and other considerations may become apparent through browsing metadata catalogs. GIS applications must by definition be sensitive to context, since they represent unique locations on the Earth’s surface.

This initial discussion is followed in [Chapter 3](#) by an examination of the methodological background to GIS analysis. Initially we examine a number of formal methodologies and then apply ideas drawn from these to the specific case of spatial analysis. A process known by its initials, PPDAC (Problem, Plan, Data, Analysis, Conclusions) is described as a methodological framework that may be applied to a very wide range of spatial analysis problems and projects. We continue Chapter 3 with a brief look at model-building, with particular reference to the various types of GIS model that can be constructed to address geospatial problems, followed by a discussion of so-called Research-ready data (RRD).

Subsequent Chapters present the various analytical methods supported within widely available software tools. The majority of the methods described in [Chapter 4 \(Building blocks of spatial analysis\)](#) and many of those in [Chapter 6 \(Surface and field analysis\)](#) are implemented as standard facilities in modern commercial GIS packages such as [ArcGIS](#), [MapInfo](#), [Manifold](#), [TNT/Datum Workstation](#) and [Intergraph](#) (now part of Hexagon Geospatial). Many are also provided in more specialized GIS products such as [TerrSet/Idrisi](#), [GRASS](#), [QGIS](#) and [ENVI](#). Note that [GRASS](#) and [QGIS](#) (which includes GRASS in its download kit) are OpenSource. A number of software tools described and illustrated in this book have since been superseded or are no longer available, but their underlying principles and methods remain applicable to the analytical processes examined. Similarly, a number of software tools have been dramatically enhanced, often as applications-specific or sector-specific add-on packages.

In addition we discuss a number of more specialized tools, designed to address the needs of specific sectors or technical problems that are otherwise not well-supported within the core GIS packages at present. [Chapter 5](#), which focuses on statistical methods, and [Chapter 7](#) and [Chapter 8](#) which address Network and Location Analysis, and Geocomputation, are less commonly supported in GIS packages, but may provide loose- or close-coupling with such systems, depending upon the application area. In all instances we provide examples and commentary on software tools that are readily available. The final Chapter ([Chapter 9](#)) addresses issues associated with so-called Big Data and AI - this is clearly an area that is developing extremely rapidly and is expected to result in major changes to the way the GIS industry operates and the forms of Geospatial Analysis that are available.

As noted above, throughout this Guide examples are drawn from and refer to specific products — these have been selected purely as examples and are not intended as recommendations. Extensive use has also been made of tabulated information, providing abbreviated summaries of techniques and formulas for reasons of both compactness and coverage. These tables are designed to provide a quick reference to the various topics covered and are, therefore, not intended as a substitute for fuller details on the various items covered. We provide limited discussion of 2D and 3D mapping facilities, and the support for digital globe formats (e.g. [KML and KMZ](#)), which is increasingly being embedded into general-purpose and specialized data analysis toolsets. These developments confirm the trend towards integration of geospatial data and presentation layers into mainstream software systems and services, both terrestrial and planetary (see, for example, the KML images of Mars DEMs developed by Google as part of the Google Earth project).

Just as all datasets and software packages contain errors, known and unknown, so too do all books and websites, and the authors of this Guide expect that there will be errors despite our

best efforts to remove these! Some may be genuine errors or misprints, whilst others may reflect our use of specific versions of software packages and their documentation. Inevitably with respect to the latter, new versions of the packages that we have used to illustrate this Guide will have appeared even before publication, so specific examples, illustrations and comments on scope or restrictions may have been superseded. In all cases the user should review the documentation provided with the software version they plan to use, check release notes for changes and known bugs, and look at any relevant online services (e.g. user/developer forums and blogs on the web) for additional materials and insights.

The web version of this Guide may be accessed via the associated Internet site: www.spatialanalysisonline.com. The contents and sample sections of the PDF version may also be accessed from this site. In both cases the information is regularly updated. The Internet is now well established as society's principal mode of information exchange and most GIS users are accustomed to searching for material that can easily be customized to specific needs. Our objective for such users is to provide an independent, reliable and authoritative first port of call for conceptual, technical, software and applications material that addresses the panoply of new user requirements.

Spatial analysis, GIS and software tools

Our objective in producing this Guide is to be comprehensive in terms of concepts and techniques (but not necessarily exhaustive), representative and independent in terms of software tools, and above all practical in terms of application and implementation. However, we believe that it is no longer appropriate to think of a standard, discipline-specific textbook as capable of satisfying every kind of new user need. Accordingly, an innovative feature of our approach here is the range of formats and channels through which we disseminate the material.

Given the vast range of spatial analysis techniques that have been developed over the past half century many topics can only be covered to a limited depth, whilst others have been omitted because they are not implemented in current mainstream GIS products. This is a rapidly changing field and increasingly GIS packages are including analytical tools as standard built-in facilities or as optional *toolsets*, *add-ins* or *analysts*. In many instances such facilities are provided by the original software suppliers (commercial vendors or collaborative non-commercial development teams) whilst in other cases facilities have been developed and are provided by third parties. Many products offer software development kits (SDKs), programming languages and language support, scripting facilities and/or special interfaces for developing one's own analytical tools or variants.

In addition, a wide variety of web-based or web-deployed tools have become available, enabling datasets to be analyzed and mapped, including dynamic interaction and drill-down capabilities, without the need for local GIS software installation. These tools include the widespread use of web-based Java, Javascript, AJAX and HTML5 applications, and interactive Virtual Globe explorers, some of which are described in this Guide. They provide an illustration of the direction that many toolset and service providers are taking.

Throughout this Guide there are numerous examples of the use of software tools that facilitate geospatial analysis. In addition, some subsections of the Guide and the software section of the accompanying website, provide summary information about such tools and links to their suppliers. Commercial software products rarely provide access to source code or full details of

the algorithms employed. Typically they provide references to books and articles on which procedures are based, coupled with online help and “white papers” describing their parameters and applications. This means that results produced using one package on a given dataset can rarely be exactly matched to those produced using any other package or through hand-crafted coding. There are many reasons for these inconsistencies including: differences in the software architectures of the various packages and the algorithms used to implement individual methods; errors in the source materials or their interpretation; coding errors; inconsistencies arising out of the ways in which different GIS packages model, store and manipulate information; and differing treatments of special cases (e.g. missing values, boundaries, adjacency, obstacles, distance computations etc.).

Non-commercial packages sometimes provide source code and test data for some or all of the analytical functions provided, although it is important to understand that “non-commercial” often does not mean that users can download the full source code. Source code greatly aids understanding, reproducibility and further development. Such software will often also provide details of known bugs and restrictions associated with functions — although this information may also be provided with commercial products it is generally less transparent. In this respect non-commercial software may meet the requirements of scientific rigor more fully than many commercial offerings, but is often provided with limited documentation, training tools, cross-platform testing and/or technical support, and thus is generally more demanding on the users and system administrators. In many instances open source and similar not-for-profit GIS software may also be less generic, focusing on a particular form of spatial representation (e.g. a grid or raster spatial model). Like some commercial software, it may also be designed with particular application areas in mind, such as addressing problems in hydrology or epidemiology.

The process of selecting software tools encourages us to ask: (i) “what is meant by geospatial analysis techniques?” and (ii) “what should we consider to be GIS software?” To some extent the answer to the second question is the simpler, if we are prepared to be guided by self-selection. For our purposes we focus principally on products that claim to provide geographic information systems capabilities, supporting at least 2D mapping (display and output) of raster (grid based) and/or vector (point/line/polygon based) data, with a minimum of basic map manipulation facilities. We concentrate our review on a number of the products most widely used or with the most readily accessible analytical facilities. This leads us beyond the realm of pure GIS. For example: we use examples drawn from packages that do not directly provide mapping facilities (e.g. the now rather outdated software called [Crimestat](#)) but which provide input and/or output in widely used GIS map-able formats; products that include some mapping facilities but whose primary purpose is spatial or spatio-temporal data exploration and analysis (e.g. GS+, [GeoDa](#), [PySal](#)); and products that are general- or special-purpose analytical engines incorporating mapping capabilities (e.g. [MATLab](#) with the Mapping Toolbox, [WinBUGS](#)) — for more details on these and other example software tools, please see the website page: www.spatialanalysisonline.com/software.html

The more difficult of the two questions above is the first — what should be considered as “geospatial analysis”? In conceptual terms, the phrase identifies the subset of techniques that are applicable when, as a minimum, data can be referenced on a two-dimensional frame and relate to terrestrial activities. The results of geospatial analysis will change if the location or extent of the frame changes, or if objects are repositioned within it: if they do not, then

“everywhere is nowhere”, location is unimportant, and it is simpler and more appropriate to use conventional, *aspatial*, techniques.

Many GIS products apply the term (geo)spatial analysis in a very narrow context. In the case of vector-based GIS this typically means operations such as: map overlay (combining two or more maps or map layers according to predefined rules); simple buffering (identifying regions of a map within a specified distance of one or more features, such as towns, roads or rivers); and similar basic operations. This reflects (and is reflected in) the use of the term *spatial analysis* within the Open Geospatial Consortium ([OGC](#)) “simple feature specifications” (see further [Table 4-2](#)). For raster-based GIS, widely used in the environmental sciences and [remote sensing](#), this typically means a range of actions applied to the grid cells of one or more maps (or images) often involving filtering and/or algebraic operations (*map algebra*). These techniques involve processing one or more raster layers according to simple rules resulting in a new map layer, for example replacing each cell value with some combination of its neighbors’ values, or computing the sum or difference of specific attribute values for each grid cell in two matching raster datasets. Descriptive statistics, such as cell counts, means, variances, maxima, minima, cumulative values, frequencies and a number of other measures and distance computations are also often included in this generic term “spatial analysis”.

However, at this point only the most basic of facilities have been included, albeit those that may be the most frequently used by the greatest number of GIS professionals. To this initial set must be added a large variety of statistical techniques (descriptive, exploratory, explanatory and predictive) that have been designed specifically for spatial and spatio-temporal data. Today such techniques are of great importance in social and political sciences, despite the fact that their origins may often be traced back to problems in the environmental and life sciences, in particular ecology, geology and epidemiology. It is also to be noted that spatial statistics is largely an observational science (like astronomy) rather than an experimental science (like agronomy or pharmaceutical research). This aspect of geospatial science has important implications for analysis, particularly the application of a range of statistical methods to spatial problems.

Limiting the definition of geospatial analysis to 2D mapping operations and spatial statistics remains too restrictive for our purposes. There are other very important areas to be considered. These include: surface analysis—in particular analyzing the properties of physical surfaces, such as gradient, aspect and visibility, and analyzing surface-like data “fields”; network analysis—examining the properties of natural and man-made networks in order to understand the behavior of flows within and around such networks; and locational analysis. GIS-based network analysis may be used to address a wide range of practical problems such as route selection and facility location, and problems involving flows such as those found in hydrology. In many instances location problems relate to networks and as such are often best addressed with tools designed for this purpose, but in others existing networks may have little or no relevance or may be impractical to incorporate within the modeling process. Problems that are not specifically network constrained, such as new road or pipeline routing, regional warehouse location, mobile phone mast positioning, pedestrian movement or the selection of rural community health care sites, may be effectively analyzed (at least initially) without reference to existing physical networks. Locational analysis “in the plane” is also applicable where suitable network datasets are not available, or are too large or expensive to be utilized, or where

the location algorithm is very complex or involves the examination or simulation of a very large number of alternative configurations.

A further important aspect of geospatial analysis is visualization (or *geovisualization*) — the use, creation and manipulation of images, maps, diagrams, charts, 3D static and dynamic views, high resolution satellite imagery and digital globes, and their associated tabular datasets (see further [Slocum et al., 2008](#), [Dodge et al., 2008](#), [Longley et al., 2015](#), Ch12). For early insights into how some of these developments may be applied, see Andrew Hudson-Smith (2008) “Digital Geography: Geographic visualization for urban environments” and Martin Dodge and Rob Kitchin’s earlier “[Atlas of Cyberspace](#)” which is available as a free downloadable document. A Working Paper on this topic from the Centre for Advanced Spatial Analysis ([CASA](#)) can be found here:

https://discovery.ucl.ac.uk/1436767/2/CASAWorkingPaper_paper190.pdf

GIS packages and web-based services increasingly incorporate a range of such tools, providing static or rotating views, draping images over 2.5D surface representations, providing animations and fly-throughs, dynamic linking and brushing and spatio-temporal visualizations. This latter class of tools has been, until recently, the least developed, reflecting in part the limited range of suitable compatible datasets and the limited set of analytical methods available, although this picture is changing rapidly. One recent example is the availability of image time series from [NASA’s Earth Observation](#) Satellites, yielding vast quantities of data on a daily basis (e.g. [Aqua mission](#), commenced 2002; [Terra mission](#), commenced 1999).

Geovisualization is the subject of ongoing research by the International Cartographic Association (ICA), [Commission on Visual Analytics](#), who have organized a series of workshops and publications addressing developments in geovisualization, notably with a cartographic focus.

As datasets, software tools and processing capabilities develop, 3D geometric and photo-realistic visualization are becoming a *sine qua non* of modern geospatial systems and services — see [Andy Hudson-Smith’s “Digital Urban” blog](#) for a regularly updated commentary on this field. We expect to see an explosion of tools and services and datasets in this area over the coming years — many examples are included as illustrations in this Guide. Other examples readers may wish to explore include: the static and dynamic visualizations at [3DNature](#) and similar sites; the 2D and 3D [Atlas of Switzerland](#); Urban 3D modeling programmes such as [CityGML](#); and the integration of GIS technologies and data with digital globe software, e.g. data from [Digital Globe \(Maxar\)](#), and Earth-based frameworks such as Google Earth, Microsoft Virtual Earth, NASA Worldwind and Edushi (Chinese). There are also automated translators between GIS packages such as [ArcGIS](#) and Digital Earth models (see for example the now withdrawn, Arc2Earth).

These novel visualization tools and facilities augment the core tools utilized in spatial analysis throughout many parts of the analytical process: exploration of data; identification of patterns and relationships; construction of models; dynamic interaction with models; and communication of results — see, for example, the work of the city of Portland, Oregon, who have used 3D visualization to communicate the results of zoning, crime analysis and other key local variables to the public; and with Covid at the forefront of much work and reporting from 2020 onward, the US interactive Covid mapping project (now ended). Another example is the

3D visualizations provided as part of the web-accessible [London Air Quality](#) network. These are designed to enable:

- users to visualize air pollution in the areas that they work, live or walk
- transport planners to identify the most polluted parts of London
- urban planners to see how building density affects pollution concentrations in the City and other high density areas, and
- students to understand pollution sources and dispersion characteristics

Physical 3D models and hybrid physical-digital models have also been developed and applied to practical analysis problems. For example: 3D physical models constructed from plaster, wood, paper and plastics have been used for many years in architectural and engineering planning projects; hybrid sand tables are used to help firefighters in California visualize the progress of wildfires (see Figure 1-1A, below and tools from companies like [SimTable](#); very large sculptured solid terrain models are being used for educational purposes, to assist land use modeling programmes, and to facilitate participatory 3D modeling in less-developed communities ([P3DM](#)); and 3D digital printing technology is being used to generate 3D landscapes and cityscapes from GIS, CAD and/or VRML files with planning, security, architectural, archaeological and geological applications (see Figure 1-1B, below and the websites of [3D Systems](#) and [Stratasys](#) for more details). To create large landscape models multiple individual prints, which are typically only around 20cm x 20cm x 5cm, are made, in much the same manner as raster file mosaics.

Figure 1-1A: 3D Physical GIS models: Sand-in-a-box model, Albuquerque, USA



Figure 1-1B: 3D Physical GIS models: 3D GIS printing



GIS software, notably in the commercial sphere, is driven primarily by demand and applicability, as manifest in willingness to pay. Hence, to an extent, the facilities available often reflect commercial and resourcing realities (including the development of improvements in processing and display hardware, and the ready availability of high quality datasets) rather than the status of development in geospatial science. Indeed, there may be many capabilities available in software packages that are provided simply because it is extremely easy for the designers and programmers to implement them, especially those employing object-oriented programming and data models. For example, a given operation may be provided for polygonal features in response to a well-understood application requirement, which is then easily enabled for other features (e.g. point sets, polylines) despite the fact that there may be no known or likely requirement for the facility.

Despite this cautionary note, for specific well-defined or *core* problems, software developers will frequently utilize the most up-to-date research on algorithms in order to improve the quality (accuracy, optimality) and efficiency (speed, memory usage) of their products. For further information on algorithms and data structures, see the online [NIST Dictionary of algorithms and data structures](#).

Furthermore, the quality, variety and efficiency of spatial analysis facilities provide an important discriminator between commercial offerings in an increasingly competitive and open market for software and spatial datasets. However, the ready availability of analysis tools does not imply that one product is necessarily better or more complete than another — it is the selection and application of *appropriate* tools in a manner that is *fit for purpose* that is important. Guidance documents exist in some disciplines that assist users in this process, e.g. Perry *et al.* (2002) dealing with ecological data analysis, and to a significant degree we hope that this Guide will assist users from many disciplines in the selection process.

Intended audience and scope

This Guide has been designed to be accessible to a wide range of readers — from undergraduates and postgraduates studying GIS and spatial analysis, to GIS practitioners and professional analysts. It is intended to be much more than a cookbook of formulas, algorithms and techniques — its aim is to provide an explanation of the key techniques of spatial analysis using examples from widely available software packages. It stops short, however, of attempting a systematic evaluation of competing software products. A substantial range of application examples are provided, but any specific selection inevitably illustrates only a small subset of the huge range of facilities available. Wherever possible, examples have been drawn from non-

academic sources, highlighting the growing understanding and acceptance of GIS technology in the commercial and government sectors.

The scope of this Guide incorporates the various spatial analysis topics included within the seminal NCGIA Core Curriculum (Goodchild and Kemp, 1990, now updated) and as such may provide a useful accompaniment to GIS Analysis courses based closely or loosely on this programme. More recently the Education Committee of the University Consortium for Geographic Information Science ([UCGIS](https://www.ucgis.org)) in conjunction with the Association of American Geographers ([AAG](https://www.aag.org)) has produced a comprehensive “Body of Knowledge” (BoK) document - see the associated website <https://www.ucgis.org/site/gis-t-body-of-knowledge>. This Guide covers materials that primarily relate to the BoK sections **CF**: Conceptual Foundations; **AM**: Analytical Methods and **GC**: Geocomputation. In the general introduction to the **AM** knowledge area the authors of the BoK summarize this component as follows:

“This knowledge area encompasses a wide variety of operations whose objective is to derive analytical results from geospatial data. Data analysis seeks to understand both first-order (environmental) effects and second-order (interaction) effects. Approaches that are both data-driven (exploration of geospatial data) and model-driven (testing hypotheses and creating models) are included. Data-driven techniques derive summary descriptions of data, evoke insights about characteristics of data, contribute to the development of research hypotheses, and lead to the derivation of analytical results. The goal of model-driven analysis is to create and test geospatial process models. In general, model-driven analysis is an advanced knowledge area where previous experience with exploratory spatial data analysis would constitute a desired prerequisite.” (BoK, p83 of the e-book version, first edition).

GIS and related software tools

The GIS software and analysis tools that an individual, group or corporate body chooses to use will depend very much on the purposes to which they will be put. There is an enormous difference between the requirements of academic researchers and educators, and those with responsibility for planning and delivery of emergency control systems or large scale physical infrastructure projects. The spectrum of products that may be described as a GIS includes (amongst others):

- mainstream GIS software packages and services, such as those from [Caliper](https://www.caliper.com) (Mapitude/TransCAD), Clark Labs ([TerrSet/IDRISI](https://www.terraserver.com)), [ESRI](https://www.esri.com) (ArcGIS Pro and related [toolboxes](#) and services), [Grass](https://grass.osgeo.org) (Opensource), [Manifold](https://www.manifold.com), and Microimages ([TNT/Datum Workstation](https://www.tnt.com))
- highly specialized, sector specific packages: for example civil engineering design and costing systems; satellite image processing systems; and utility infrastructure management systems
- transportation and logistics management systems
- civil and military control room systems
- systems for visualizing the built environment for architectural purposes, for public consultation or as part of simulated environments for interactive gaming
- land registration systems

- census data management systems
- commercial location services and Digital Earth models
- Geospatial data visualization tools
- AI-augmented geospatial data services and software tools
- Online versions of offline (desktop) GIS software

The list of software functions and applications is long and in some instances suppliers would not describe their offerings as a GIS. In many cases such systems fulfill specific operational needs, solving a well-defined subset of spatial problems and providing mapped output as an incidental but essential part of their operation. Many of the capabilities may be found in generic GIS products. In other instances a specialized package may utilize a GIS engine for the display and in some cases processing of spatial data (directly, or indirectly through interfacing or file input/output mechanisms). For this reason, and in order to draw a boundary around the present work, reference to application-specific GIS will be limited.

A number of GIS packages and related toolsets have particularly strong facilities for processing and analyzing binary, grayscale and color images. They may have been designed originally for the processing of remote sensed data from satellite and aerial surveys, but many have developed into much more sophisticated and complete GIS tools, e.g. Clark Lab's [Terrset/Idrisi](#) software; MicroImage's [TNT/Datum Workstation](#) product set; the [ERDAS](#) suite of products; and [ENVI](#) with associated packages such as [RiverTools](#). Alternatively, image handling may have been deliberately included within the original design parameters for a generic GIS package (e.g. [Manifold](#)), or simply be toolsets for image processing that may be combined with mapping tools (e.g. the [MATLab](#) Image Processing Toolbox). Whatever their origins, a central purpose of such tools has been the capture, manipulation and interpretation of image data, rather than spatial analysis *per se*, although the latter inevitably follows from the former.

In this Guide we do not provide a separate chapter on image processing, despite its considerable importance in GIS, focusing instead on those areas where image processing tools and concepts are applied for spatial analysis (e.g. surface analysis). We have adopted a similar position with respect to other forms of data capture, such as field and geodetic survey systems and data cleansing software — although these incorporate analytical tools, their primary function remains the recording and georeferencing of datasets, rather than the analysis of such datasets once stored.

For most GIS professionals, spatial analysis and associated modeling is an infrequent activity. Even for those whose job focuses on analysis the range of techniques employed tends to be quite narrow and application specific. GIS consultants, researchers and academics on the other hand are continually exploring and developing analytical techniques. For the first group and for consultants, especially in commercial environments, the imperatives of financial considerations, timeliness and corporate policy loom large, directing attention to: delivery of solutions within well-defined time and cost parameters; working within commercial constraints on the cost and availability of software, datasets and staffing; ensuring that solutions are fit for purpose/meet client and end-user expectations and agreed standards; and in some cases, meeting “political” expectations.

For the second group of users it is common to make use of a variety of tools, data and programming facilities developed in the academic sphere. Increasingly these make use of non-commercial wide-ranging spatial analysis software libraries, such as the [R-Spatial](#) project (in “R”); [PySal](#) (in “Python”); and [Splancs](#) (in “S”). Many of these tools include support for a variety of file types and formats, including those required for mapping the results (e.g. using [ESRI](#)-type shape, shp, file handling) - for more details and worked examples using R, see Lansley and Cheshire's (2016) [Introductory Guide](#) downloadable as a PDF.

Sample software products

The principal products we have included in this latest edition of the Guide are included on the accompanying website's [software page](#). Many of these products are free whilst others are available (at least in some form) for a small fee for all or selected groups of users. Others are licensed at varying per user prices, from a few hundred to over a thousand US dollars per user. Our tests and examples have largely been carried out using desktop/Windows versions of these software products. Different versions that support Unix-based operating systems and more sophisticated back-end database engines have not been utilized. In the context of this Guide we do not believe these selections affect our discussions in any substantial manner, although such issues may have performance and systems architecture implications that are extremely important for many users. Increasingly online services are being offered that provide access to very large datasets, data mining and AI-driven analytical tools; brief discussion of the opportunities and issues relating to some of these services is provided in the final Chapter of this book. [OGC](#) compliant software products are listed on the [OGC](#) resources web page: https://portal.ogc.org/public_ogc/compliance/compliant.php

To quote from the latest [OGC](#) statement: “*The OGC Compliance Program is a certification process that ensures organizations’ solutions are compliant with OGC Standards. It is a universal credential that allows agencies, industry, and academia to better integrate their solutions. OGC compliance provides confidence that a product will seamlessly integrate with other compliant solutions regardless of the vendor that created them*”

Software performance

Suppliers should be able to provide advice on performance issues (e.g. see the [ESRI](#) web site, "Services" area for relevant material relating to their products) and in some cases such information is provided within product Help files (e.g. see the Performance Tips section within the [Manifold](#) GIS support section). Some analytical tasks are very processor- and memory-hungry, particularly as the number of elements involved increases. For example, vector overlay and buffering is relatively fast with a few objects and layers, but slows appreciably as the number of elements involved increases. This increase is generally at least linear with the number of layers and features, but for some problems grows in a highly non-linear (i.e. geometric) manner. Offline (desktop) software like [Manifold](#) has, in recent years, been dramatically improved in terms of data loading and processing speed, notably by using multiple cores and processor units, in particular, NVIDIA GPUs.

Many optimization tasks, such as optimal routing through networks or trip distribution modeling, are known to be extremely hard or impossible to solve optimally and methods to achieve a best solution with a large dataset can take a considerable time to run (see our discussion of [Algorithms and computational complexity theory](#) later in this Guide for a fuller

discussion of this topic). Similar problems exist with the processing and display of raster files, especially large images or sets of images. Geocomputational methods, some of which are beginning to appear within GIS packages and related toolsets, are almost by definition computationally intensive. This certainly applies to large-scale ([Monte Carlo](#)) simulation models, cellular automata and agent-based models and some raster-based optimization techniques, especially where modeling extends into the time domain.

A frequent criticism of GIS software is that it is over-complicated, resource-hungry and requires specialist expertise to understand and use. Such criticisms are often valid and for many problems it may prove simpler, faster and more transparent to utilize specialized tools for the analytical work and draw on the strengths of GIS in data management and mapping to provide input/output and visualization functionality. Example approaches include: (i) using high-level programming facilities within a GIS (e.g. macros, scripts, VBA, [Python](#)) – many add-ins are developed in this way; (ii) using wide-ranging programmable spatial analysis software libraries and toolsets that incorporate GIS file reading, writing and display, such as the [R-Spatial](#) and [PySal](#) projects noted earlier; (iii) using general purpose data processing toolsets, e.g. [MATLab](#), Excel, [Matplotlib](#), Numeric Python ([Numpy](#)), Cython ("an optimizing static compiler for both the [Python](#) programming language and the extended Cython programming language (based on Pyrex). It makes writing C extensions for Python as easy as Python itself" - see further <https://cython.org/>), and other libraries from [Enthought](#); or (iv) directly utilizing mainstream programming languages (e.g. Java, C++). The advantage of these approaches is control and transparency, the disadvantages are that software development is never trivial, is often subject to frustrating and unforeseen delays and errors, and generally requires ongoing maintenance. In some instances analytical applications may be well-suited to parallel or grid-enabled processing – as for example is the case with [GWR](#) (see Harris *et al.*, 2006). In some instances many of these issues are effectively 'outsourced' to a geospatial services provider, with datasets, mapping and analytical toolsets delivered through a single, user-friendly interface, with all the pros and cons of such packaged offerings.

At present there are no standardized tests for the quality, speed and accuracy of GIS procedures. It remains the buyer's and user's responsibility and duty to evaluate the software or service they wish to use for the specific task at hand, and by systematic controlled tests or by other means establish that the product and facility within that product they choose to use is truly fit for purpose — *caveat emptor*! Details of how to obtain many of these products are provided on the [software page](#) of the website that accompanies this book. The list maintained on Wikipedia is also a useful source of information and links, although is far from being complete or independent. A number of trade magazines and websites provide ad hoc reviews of GIS software and service offerings, especially new releases, although coverage of analytical functionality may be limited.

Suggested reading

There are numerous excellent modern books on GIS and spatial analysis, although few address software facilities and developments. Hypertext links are provided here, and throughout the text where they are cited, to the more recent publications and web resources listed.

As a background to this Guide any readers unfamiliar with GIS are encouraged to first tackle "Geographic Information Science and Systems" (GISSc) by [Longley et al.](#) (2015). GISSc

seeks to provide a comprehensive and highly accessible introduction to the subject as a whole. The GB Ordnance Survey's "[GIS pages](#)" also provides an excellent brief introduction to GIS and its applications.

Some of the basic mathematics and statistics of relevance to GIS analysis is covered in Dale (2005) and Allan (2004). For detailed information on datums and map projections, see Iliffe and Lott (2008). Useful online resources for those involved in data analysis, particularly with a statistical content, include the [StatsRef website](#) and the [e-Handbook of Statistical Methods](#) produced by the US National Institute on Standards and Technology, NIST). The more informally produced set of articles on statistical topics provided under the Wikipedia umbrella are also an extremely useful resource. These sites, and the mathematics reference site, [Mathworld](#), are referred to (with hypertext links) at various points throughout this document. For those who find mathematics and statistics something of a mystery, [de Smith \(2018\)](#) and Bluman (2003) provide useful starting points. For guidance on how to avoid the many pitfalls of statistical data analysis readers are recommended the material in the classic work by Huff (1993) "How to lie with statistics", and the 2008 book by Blastland and Dilnot "The tiger that isn't".

A relatively new development has been the increasing availability of out-of-print published books, articles and guides as free downloads in PDF format. These include: the series of 59 short guides published under the [CATMOG](#) umbrella (Concepts and Methods in Modern Geography), published between 1975 and 1995, most of which are now available at the [QMRG](#) website (a full list of all the guides is provided at the end of this book); the [Atlas of Cyberspace](#) by Dodge and Kitchin; and [Fractal Cities](#), by Batty and Longley.

Undergraduates and MSc programme students will find Burrough and McDonnell (1998, 2015) provides excellent coverage of many aspects of geospatial analysis, especially from an environmental sciences perspective. Valuable guidance on the relationship between spatial process and spatial modeling may be found in Cliff and Ord (1981) and Bailey and Gatrell (1995). The latter provides an excellent introduction to the application of statistical methods to spatial data analysis. O'Sullivan and Unwin (2010, 2nd ed.) is a more broad-ranging book covering the topic the authors describe as "Geographic Information Analysis". This work is best suited to advanced undergraduates and first year postgraduate students. In many respects a deeper and more challenging work is Haining's (2003) "Spatial Data Analysis — Theory and Practice". This book is strongly recommended as a companion to the present Guide for postgraduate researchers and professional analysts involved in using GIS in conjunction with statistical analysis.

However, these authors do not address the broader spectrum of geospatial analysis and associated modeling as we have defined it. For example, problems relating to networks and location are often not covered and the literature relating to this area is scattered across many disciplines, being founded upon the mathematics of graph theory, with applications ranging from electronic circuit design to computer networking and from transport planning to the design of complex molecular structures. Useful books addressing this field include Miller and Shaw (2001) "Geographic Information Systems for Transportation" (especially Chapters 3, 5 and 6), and Rodrigue *et al.* (2006) "The geography of transport systems" (see further: <https://transportgeography.org/>).

As companion reading on these topics for the present Guide we suggest the two volumes from the Handbooks in Operations Research and Management Science series by Ball *et al.* (1995): “Network Models”, and “Network Routing”. These rather expensive volumes provide collections of reviews covering many classes of network problems, from the core optimization problems of shortest paths and arc routing (e.g. street cleaning), to the complex problems of dynamic routing in variable networks, and a great deal more besides. This is challenging material and many readers may prefer to seek out more approachable material, available in a number of other books and articles, e.g. Ahuja *et al.* (1993), Mark Daskin’s excellent book “Network and Discrete Location” (1995) and the earlier seminal works by Haggett and Chorley (1969), and Scott (1971), together with the widely available online materials accessible via the Internet. Final recommendations here are Stephen Wise’s excellent GIS Basics (2002) and Worboys and Duckham (2004) which address GIS from a computing perspective. Both these volumes covers many topics, including the central issues of data modeling and data structures, key algorithms, system architectures and interfaces.

Many recent books described as covering (geo)spatial analysis are essentially edited collections of papers or brief articles. As such most do not seek to provide comprehensive coverage of the field, but tend to cover information on recent developments, often with a specific application focus (e.g. health, transport, archaeology). The latter is particularly common where these works are selections from sector- or discipline-specific conference proceedings, whilst in other cases they are carefully chosen or specially written papers. Classic amongst these is Berry and Marble (1968) “Spatial Analysis: A reader in statistical geography”. More recent examples include “GIS, Spatial Analysis and Modeling” edited by Maguire, Batty and Goodchild (2005), and the excellent (but costly) compendium work “The SAGE handbook of Spatial Analysis” edited by Fotheringham and Rogerson (2008).

A second category of companion materials to the present work is the extensive product-specific documentation available from software suppliers. Some of the online help files and product manuals are excellent, as are associated example data files, tutorials, worked examples and white papers (see for example, [ESRI’s What is GIS?](#)), which provides a wide-ranging guide to GIS. In many instances we utilize these to illustrate the capabilities of specific pieces of software and to enable readers to replicate our results using readily available materials. In addition some suppliers, notably [ESRI](#), have a substantial publishing operation, including more general (i.e. not product specific) books of relevance to the present work. Amongst their publications we strongly recommend the “ESRI Guide to GIS Analysis Volume 1: Geographic patterns and relationships” (1999) by Andy Mitchell, which is full of valuable tips and examples. This is a basic introduction to GIS Analysis, which he defines in this context as “a process for looking at geographic patterns and relationships between features”. Mitchell’s Volume 2 (July 2005) covers more advanced techniques of data analysis, notably some of the more accessible and widely supported methods of spatial statistics, and is equally highly recommended. A number of the topics covered in his Volume 2 also appear in this Guide. David Allen has produced a tutorial book and DVD (GIS Tutorial II: Spatial Analysis Workbook) to go alongside Mitchell’s volumes, and these are obtainable from ESRI Press. For a perspective on modern applications of GIS and geospatial analysis see Esri’s [GIS For Science series](#) (Volume 3 is the latest and Chapter 7 focuses on geospatial analysis and AI, also referred to now as GeoAI) and the recently published collection of edited chapters brought together as the “Handbook of Geospatial Artificial Intelligence” (2024, Song Gao *et al.*, with a Foreword

by Prof Mike Goodchild). Those considering using Open Source software may wish to investigate the books by Neteler and Mitasova (2008), Tyler Mitchell (2005) and Sherman (2008).

In parallel with the increasing range and sophistication of spatial analysis facilities to be found within GIS packages, there has been a major change in spatial analytical techniques. In large measure this has come about as a result of technological developments and the related availability of software tools and detailed publicly available datasets. One aspect of this has been noted already — the move towards network-based location modeling where in the past this would have been unfeasible. More general shifts can be seen in the move towards local rather than simply global analysis, for example in the field of exploratory data analysis; in the increasing use of advanced forms of visualization as an aid to analysis and communication; in the growth of online spatial data, mapping and analysis services; in the development of a wide range of computationally intensive and simulation methods that address problems through micro-scale processes (geocomputational methods); and finally, in the recent development of AI-augmented services, many of which fit within the somewhat portmanteau term, GeoAI. These trends are addressed at many points throughout this Guide.

Geospatial analysis involves a wide range of terms used in various contexts, often borrowed from mathematics, statistics, and geography, where meanings may vary depending on the source or software. Below is a comprehensive paragraph-style explanation of selected key terminologies used in geospatial analysis, based on widely recognized sources like the Association for Geographic Information (AGI) and the Open Geospatial Consortium (OGC).

Adjacency is when two or more polygons share a common boundary or side, although this concept also applies to features on opposite sides of a shared boundary, not limited to polygons. An arc typically refers to a straight line segment connecting two nodes in a polyline or polygon but may also include curves or splines, with directed or undirected types having attributes like cost or capacity in network analysis. Artifacts are unusual, insignificant observations such as spikes in 3D plots, often caused by data collection or computation processes, sometimes appearing as ghost lines.

Aspect indicates the direction where the slope of a surface is steepest, relating directly to slope and gradient. An attribute is a data item linked with an object in a spatial database, which can be explicit (stored in tables) or implicit (computed on demand), and raster datasets usually have one attribute value per cell. Azimuth measures the horizontal direction of a vector clockwise in degrees from the positive Y-axis, akin to compass bearings, while azimuthal projection maps areas as if projecting onto a tangent plane, preserving true compass bearings.

Spatial autocorrelation describes the degree to which spatial variables vary together, positively or negatively, commonly seen in ordered datasets like time series or spatial data ordered by distance bands; although it implies association, it does not prove causality. A cartogram distorts map geometry to represent data like population instead of land area, while choropleth maps visualize data categories across areas using color or shading.

Conflation merges data sources into one by reconciling differences, distinct from concatenation where sources overlay but remain separate. Contiguity represents the topological relationship by recording adjacent polygons by arcs, focusing on relative rather than exact geographic

locations. A curve is a one-dimensional geometric object defined by point sequences with interpolation, where LineString or polyline is a common linear subtype.

A datum refers to a reference surface model (e.g., WGS84) used for elevation or horizontal geographic referencing. DEM and DTM represent digital elevation and terrain models, respectively, while EDM denotes electronic distance measurement. EDA and ESDA refer to exploratory data analysis and exploratory spatial data analysis. Ellipsoid or spheroid describes the Earth's shape as an ellipse rotated about its minor axis.

A feature is a GIS object such as a point, line, polygon, or annotation, with geoid representing Earth's mean sea level extended under continents as a gravitational surface. Geodemographics analyze populations by locality or neighborhood type. Geospatial pertains to location relative to Earth's surface and often involves spatial data representations beyond simple maps. Geostatistics are specialized statistical methods dealing with spatial data, accounting for issues like autocorrelation, often applied in environmental and social sciences.

Geovisualization includes techniques that visualize spatial and spatio-temporal data, from 2D maps to dynamic 3D models and immersive systems. GIS-T focuses on GIS applications in transportation. GPS and DGPS are satellite-based positioning systems with DGPS providing enhanced accuracy through fixed reference stations. Gradient is a vector field of slope magnitude and aspect direction across surfaces.

Graphs are mathematical collections of vertices and edges studied in graph theory. Heuristics are problem-solving approaches aimed at finding near-optimal solutions efficiently. IID is an abbreviation for independently and identically distributed data samples. Invariance refers to spatial features maintaining properties under certain transformations. Kernel methods involve localized calculations within defined neighborhoods.

Layers are collections of geographic entities of a single type, which can be combined across types. Map algebra involves cell-based algebraic operations on raster data to produce new layers. A mashup refers to web content created by integrating multiple data sources via programmable interfaces. The minimum bounding rectangle (MBR) or minimum enclosing rectangle (MER) defines the smallest enclosing rectangular boundary for feature sets.

Planar and non-planar refer to geometries lying in a single plane or not, with planar enforced polygons having no overlaps except at boundaries; planar graphs can be drawn without edge intersections outside nodes. Pixels are the smallest image units carrying color or intensity values. A polygon is a closed planar figure defined by connected vertices and edges; if simple, it does not self-intersect, and convex if all internal lines lie within it.

Polyhedral surfaces consist of connected polygons sharing boundaries, while polylines are vertex-ordered connected line segments, with LineStrings a GIS standard curve subtype. Raster/grid data represent geographic features as a grid of discrete cells, each holding values, often referenced from the lower-left corner. Resampling adjusts raster resolutions or reduces dataset size by aggregating pixels, with coarse images containing less detail.

Rubber sheeting adjusts dataset coordinates for alignment by stretching or shrinking while preserving topology. Slope measures elevation change over horizontal distance, with the maximum slope direction called aspect; combined, they form the gradient vector. Spatial

econometrics applies econometric methods to spatial data focusing on spatial dependence and heterogeneity across locations and time.

A spheroid models Earth's shape as a slightly flattened sphere (oblate ellipsoid). SQL in GIS supports spatial queries based on relationships rather than just attributes. Surfaces are 2D geometric objects possibly with holes, while tessellations divide planes into polygons like raster squares or triangles (TIN). Triangulated irregular networks (TINs) model surfaces by connecting irregularly spaced points into non-overlapping triangles.

Topology captures spatial relationships such as connectivity and adjacency independent of exact coordinates, represented in digital data by nodes and links. Transformations convert coordinate data between systems, including affine transformations preserving parallel lines. Data transformations apply mathematical functions on datasets, with back transformations reverting to original data. Vectors represent GIS data as lines or arcs with explicit node storage.

Viewshed maps visibility regions from observation points, indicating visible raster cells. WGS84 is a major global reference system modeling Earth as a spheroid with precise flattening at poles.

Common Measures and Notation

Throughout this Guide a number of terms and associated formulas are used that are common to many analytical procedures. In this section we provide a brief summary of those that fall into this category. Others, that are more specific to a particular field of analysis, are treated within the section to which they primarily apply. Many of the measures we list will be familiar to readers, since they originate from standard single variable (univariate) statistics. For brevity we provide details of these in tabular form. In order to clarify the expressions used here and elsewhere in the text, we use the notation shown in Table 1-2. Italics are used within the text and formulas to denote variables and parameters, as well as selected terms.

The notation and symbology used in geospatial analysis and related mathematical contexts include several fundamental symbols and their meanings. The closed interval $[a,b]$ on the real line includes all values between a and b , including the endpoints. In contrast, (a,b) denotes an open interval, containing all values between a and b but excluding the endpoints. This open interval symbol should not be confused with coordinate pairs like (x,y) or function inputs like $f(x,y)$, where the meaning depends on context. In graph theory, (i,j) represents an edge connecting vertices i and j . The (x,y) pair typically represents a two-dimensional spatial coordinate in Cartesian or spherical form, with latitude and longitude ranges commonly from -90 to $+90$ and -180 to $+180$ degrees respectively. Adding a third dimension as (x,y,z) includes height, depth, or an attribute value associated with the spatial location.

A set of n values $x_1, x_2, \dots, x_n$, denoted $\{x_i\}$, represents data points or measurements, possibly continuous or ratio-scaled. An ordered set $\{X_i\}$ arranges these values so that $x_i \leq x_{i+1}$ for all i , useful in statistics. Bold uppercase letters (X) denote matrices, while lowercase bold letters (x) indicate vectors. Frequency counts of data values or classes are represented by $\{f_i\}$, with the sum of all f_i equal to the number of observations n . Probabilities $\{p_i\}$ derived from frequencies satisfy the summation to unity, $\sum p_i = 1$.

Symbol Σ denotes summation (e.g., $x_1 + x_2 + \dots + x_n$), while Π indicates multiplication across elements. The caret symbol $\hat{}$ marks an estimate of a population parameter, often called a "hat".

The tilde $\sim$ signifies a variable's distribution, for example $y \sim N(0,1)$ indicates y is normally distributed with mean 0 and standard deviation 1. The factorial symbol $!$ defines multiplicative sequences for integers, such as $x! = x(x-1)(x-2)\dots 1$. The symbol $\equiv$ means "equivalent to," while $\approx$ means "approximately equal to." The symbol $\in$ is used for set membership, expressing that x belongs to a set or interval, e.g., $x \in$. Inequality symbols $\leq$ (less than or equal to) and $\geq$ (greater than or equal to) are sometimes represented as $\leq$ and $\geq$ for compatibility with text displays.

These symbols form the concise notation backbone for representing intervals, sets, coordinate pairs, vectors, matrices, probabilities, estimations, distributions, and mathematical relationships widely used in geospatial data analysis and statistics.mathvault+2